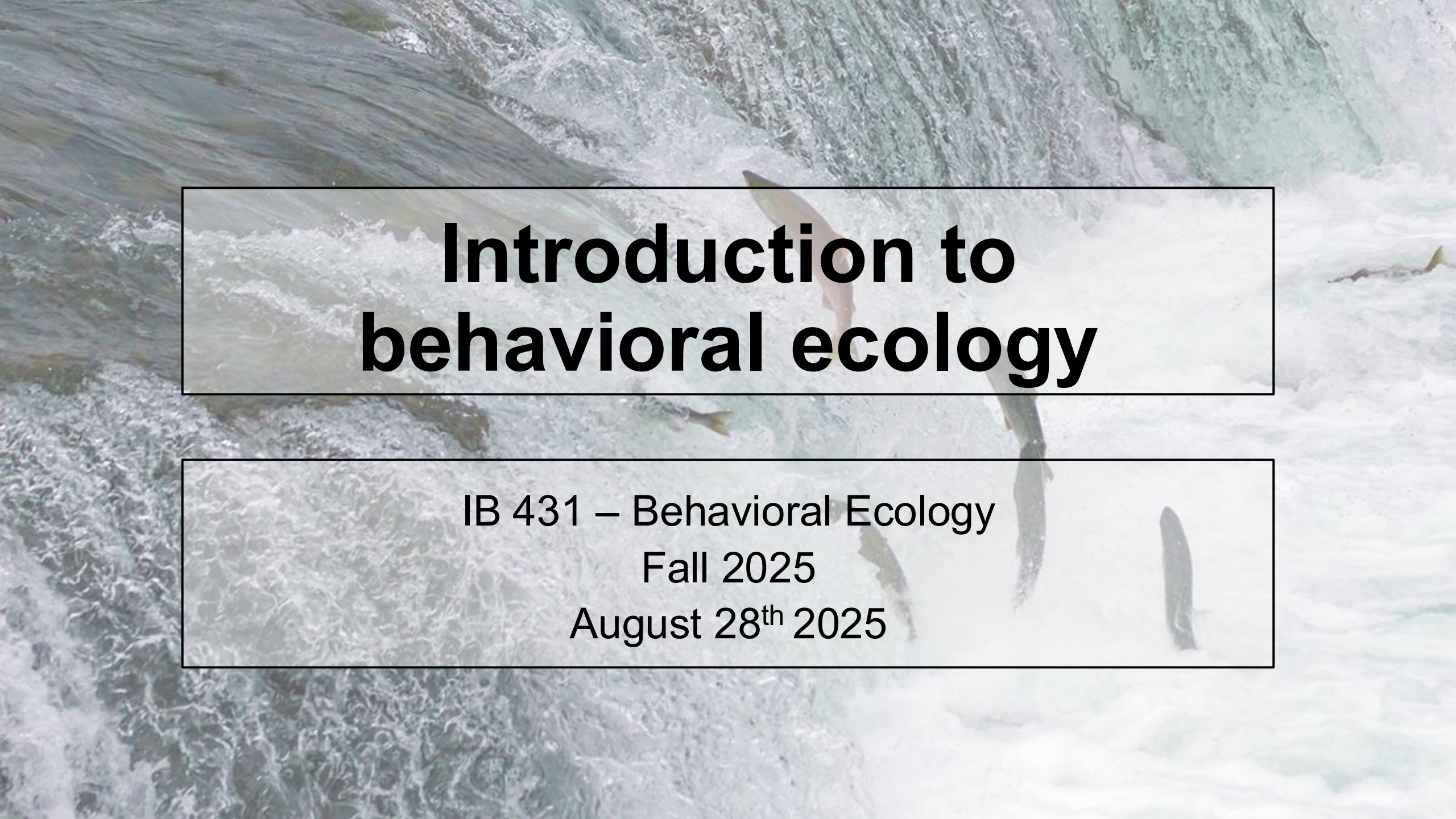


A background image showing several salmon jumping over a waterfall. The water is white and frothy from the falls, and the surrounding rocks are wet and mossy. The salmon are in various stages of their jump, with some fully airborne and others just entering the water.

Introduction to behavioral ecology

IB 431 – Behavioral Ecology
Fall 2025
August 28th 2025

Announcements

- Sign up to lead discussions!
We still need people for next week!

Outline

- I. What is behavioral ecology?
- II. Why do animals do what they do?
- III. “Good for the species” arguments
- IV. Discussing scientific articles
- V. Discuss Pope 2000

What is behavioral ecology?

- Study of the evolutionary basis of animal behavior in the context of ecological interactions



What is behavioral ecology?

- Study of the evolutionary basis of animal behavior in the context of ecological interactions
 - Social behavior
 - Mutualism
 - Predator/prey
 - Foraging
 - Pathogens / parasitism
 - Competition / territoriality



Interest in animal behavior spans across history and cultures

- Cave paintings, Stone Age (~4000 – 2000 BCE)



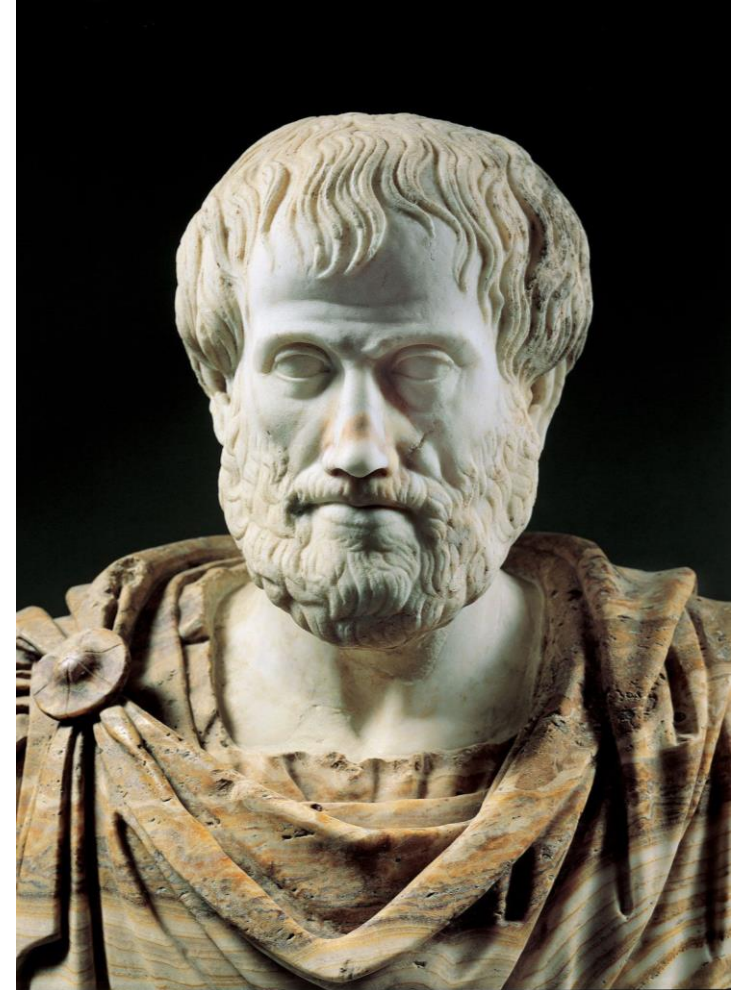
Interest in animal behavior spans across history and cultures

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- **Zhou Dynasty (5th century BCE)**



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- **Maya civilization (3rd century BCE)**

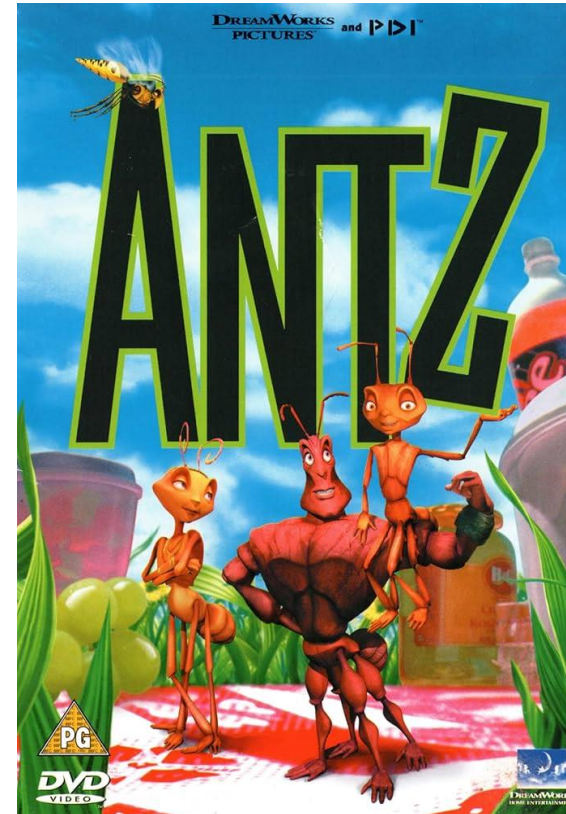
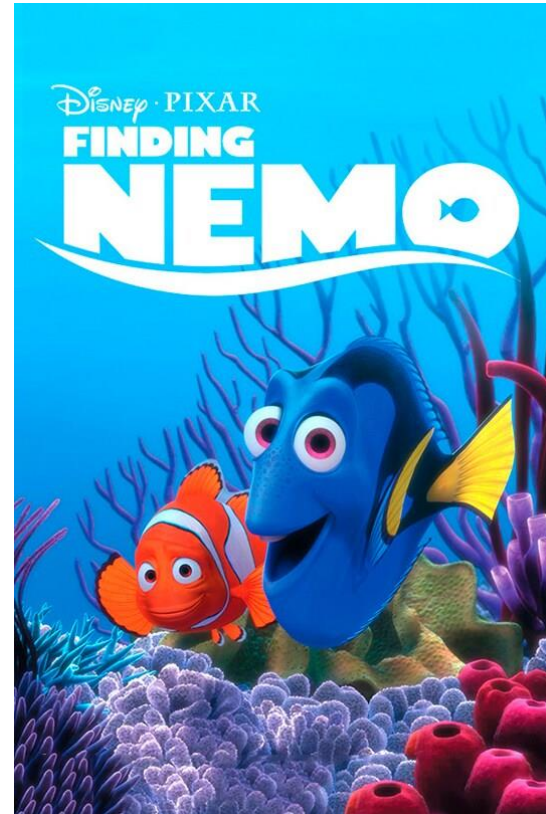
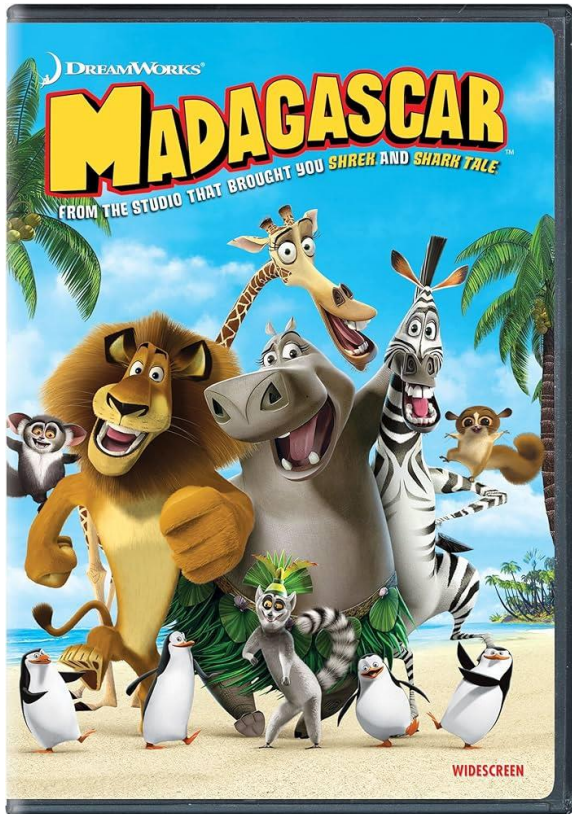


Interest in animal behavior spans across history and cultures

- Cave paintings, Stone Age (~4000 – 2000 BCE)
- Zhou Dynasty (5th century BCE)
- Aristotle, *Historia animalium* (4th century BCE)
- Maya civilization (3rd century BCE)
- **al-Jahiz, *Kitab Al-Hayawan* (Book of Animals) (8th century CE)**



This interest has continued
in modern popular culture



Variation

- Genetic
- Phenotypic
- Environmental



Wall lizards (*Podarcis* sp.)

Variation

- Genetic
- Phenotypic
- Environmental



Interactions

- With abiotic environment
- Within individuals (i.e., phenotypic correlations)
- Within species
- Across species

Wall lizards (*Podarcis* sp.)



Outline

- I. What is behavioral ecology?
- II. **Why do animals do what they do?**
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Why do animals do what they do?



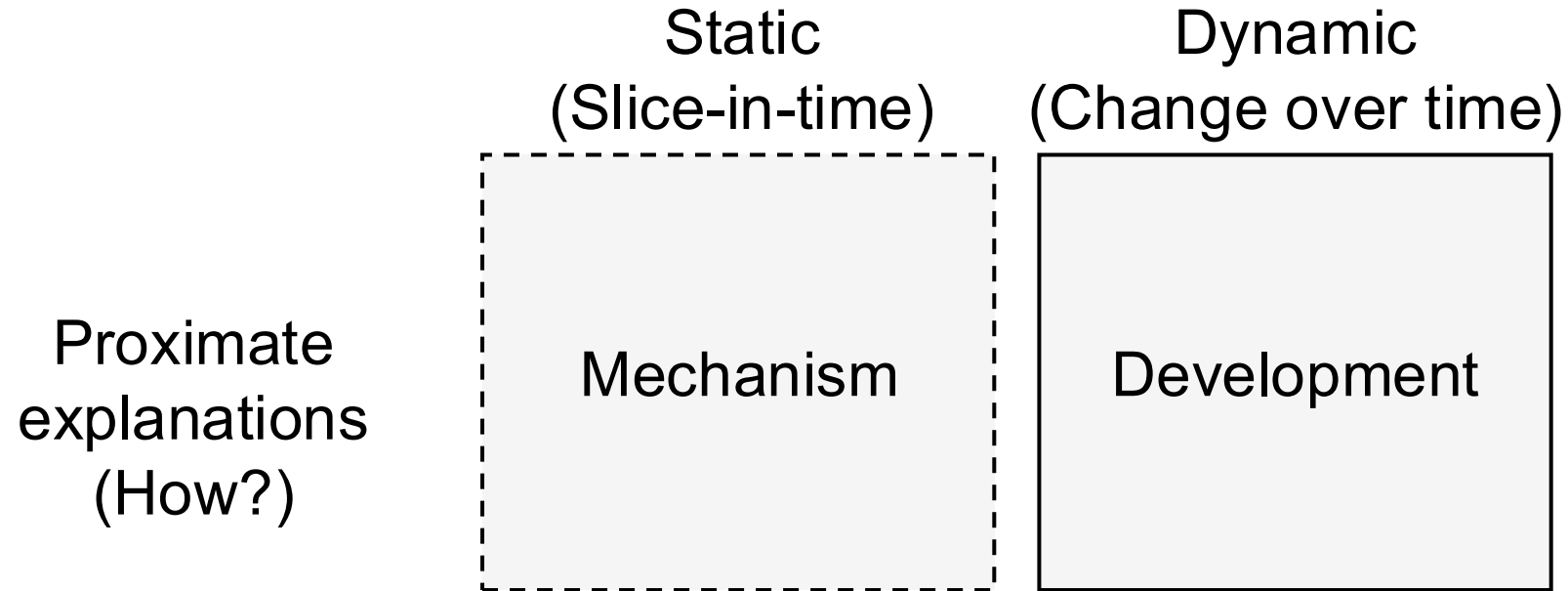
Why do animals do what they do?

- Genes / gene expression
- Hormones
- Brain / neural mechanisms
- Learned behaviors
- Responding to other organisms
- Acquire resources
- Drive to reproduce
- Ancestors were exposed to conditions favoring a behavior
- Climatic / weather conditions

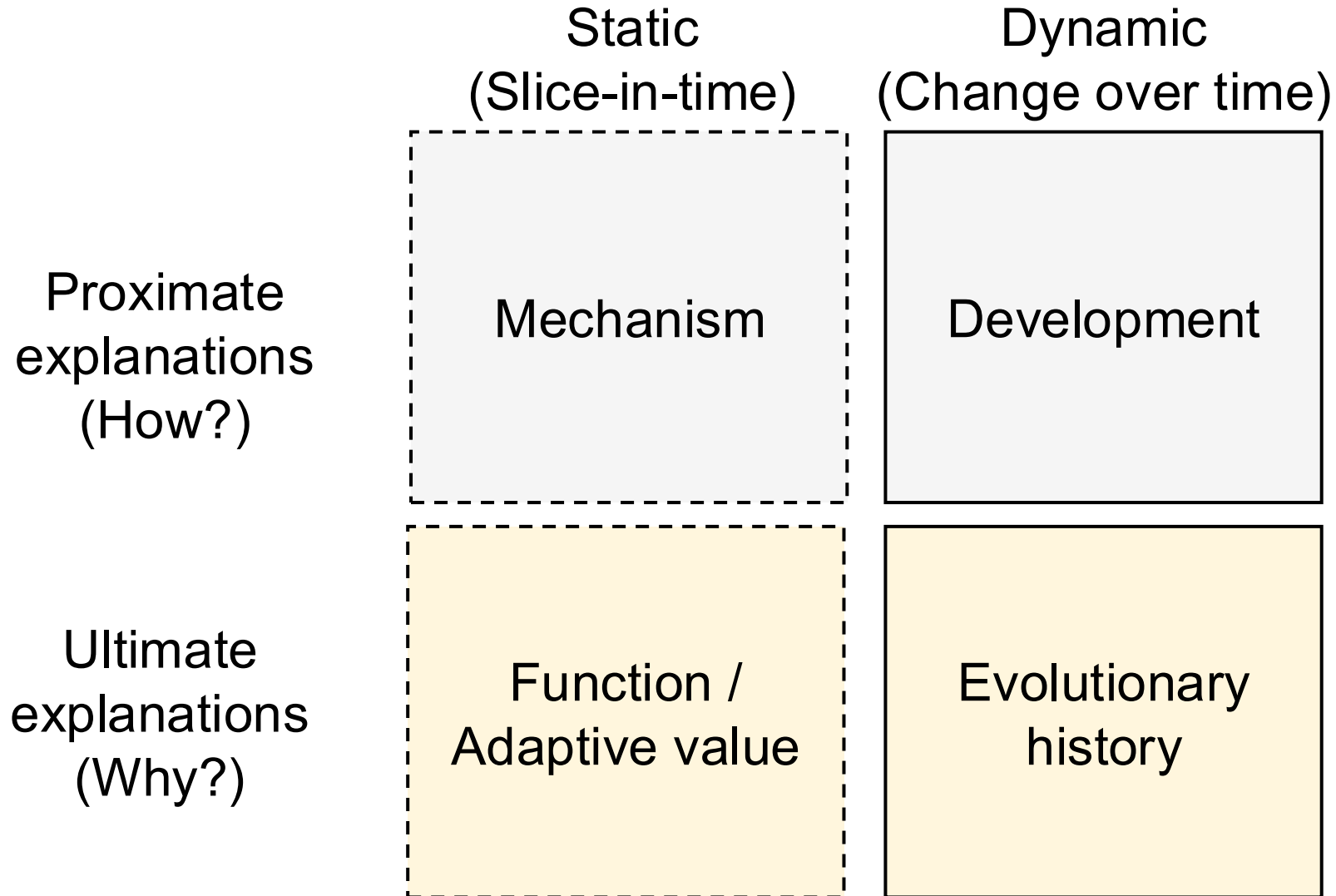


Tinbergen's four questions

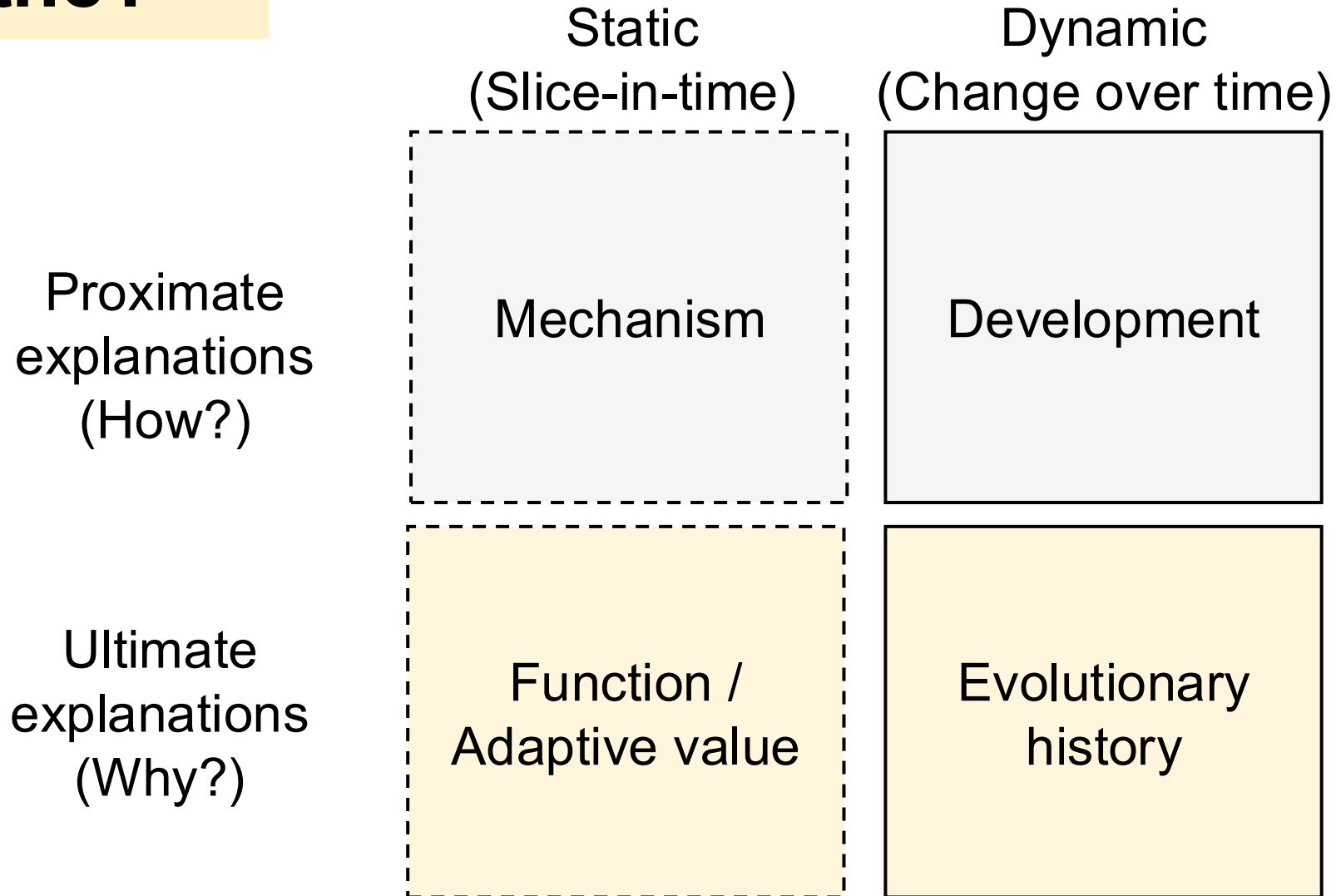
Tinbergen's four questions



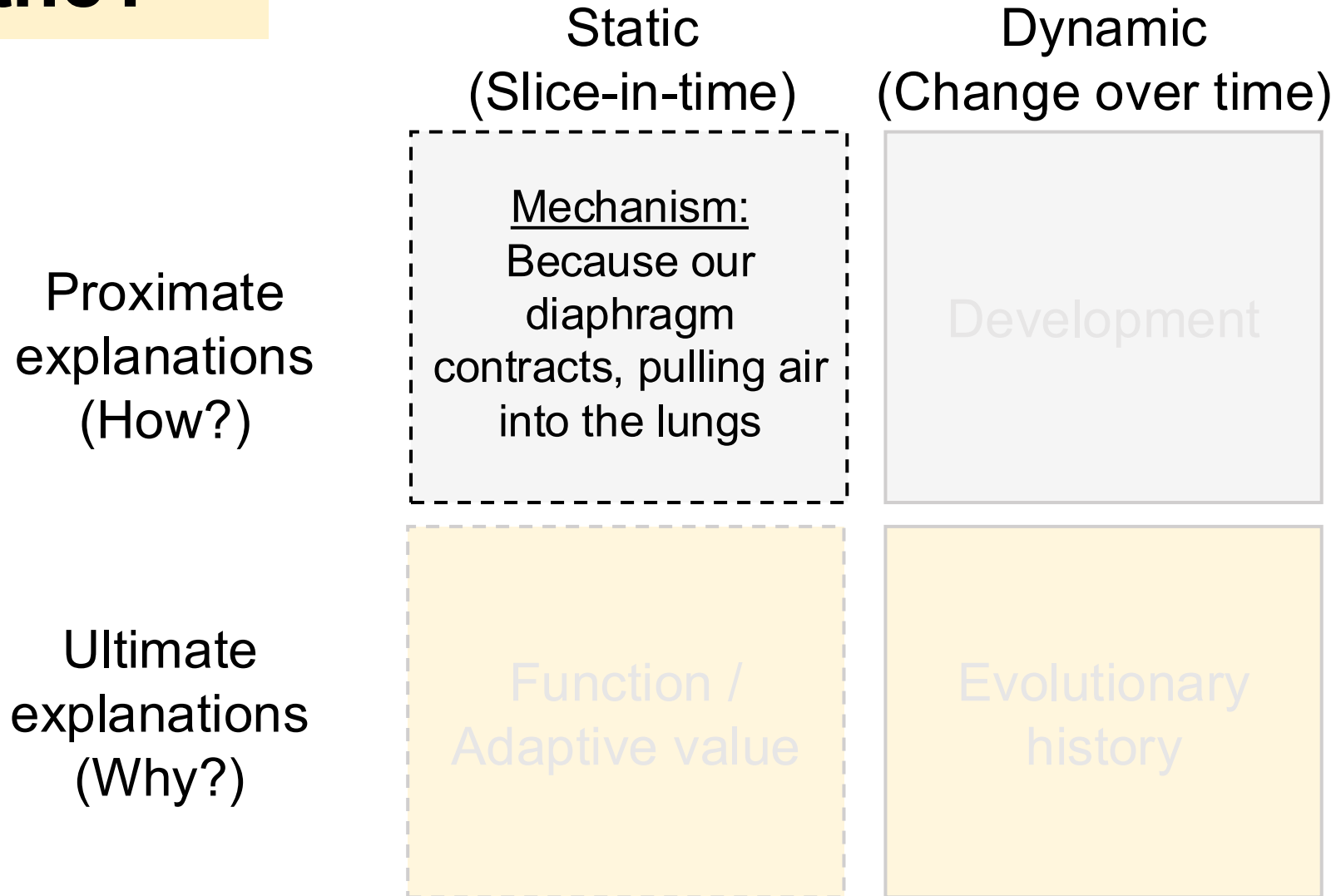
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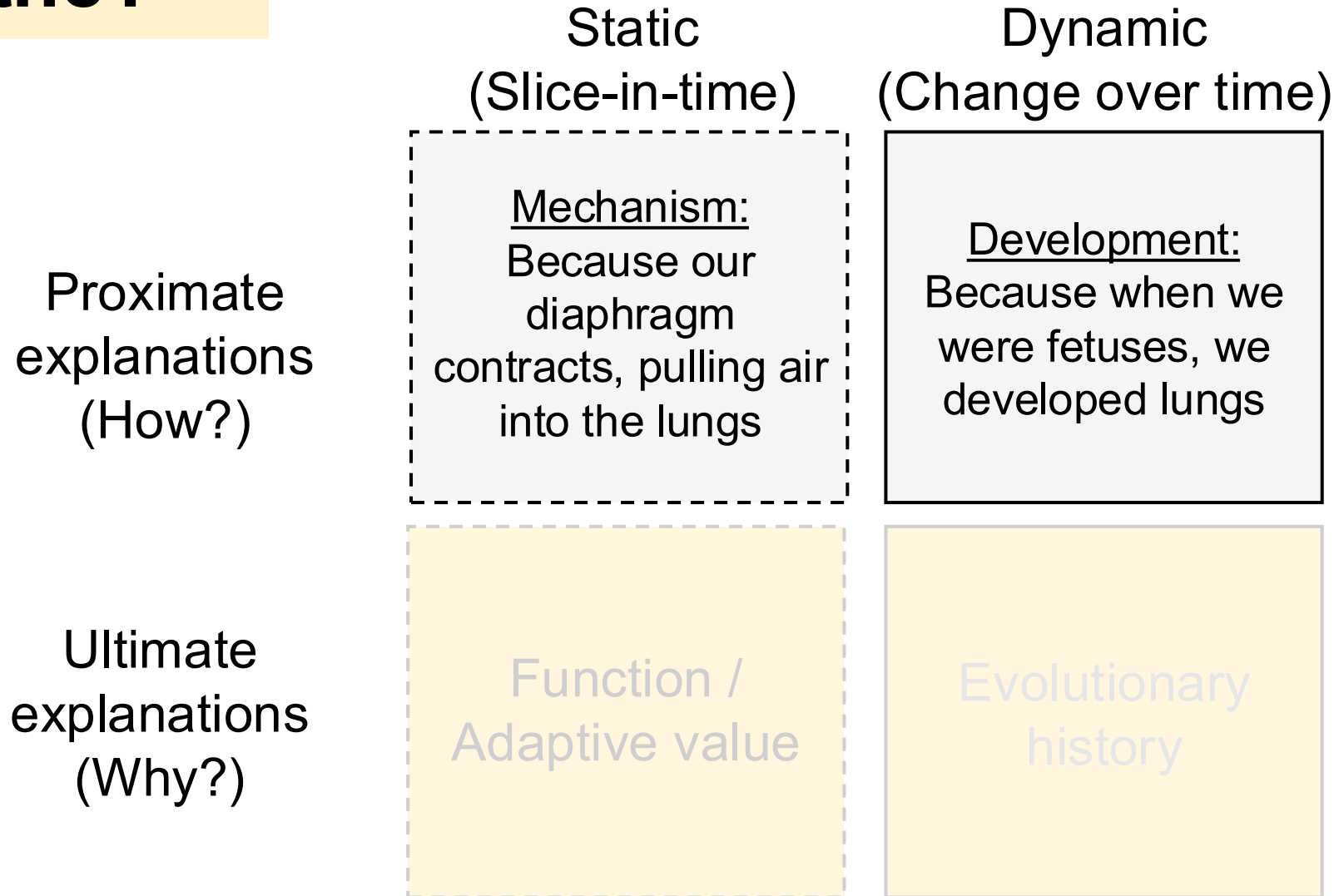
Why do we breathe?



Why do we breathe?



Why do we breathe?



Why do we breathe?

	Static (Slice-in-time)	Dynamic (Change over time)
Proximate explanations (How?)	<u>Mechanism:</u> Because our diaphragm contracts, pulling air into the lungs	<u>Development:</u> Because when we were fetuses, we developed lungs
Ultimate explanations (Why?)	<u>Function:</u> Because it brings in oxygen which keeps us alive	Evolutionary history

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Ultimate explanations (Why?)	<u>Function:</u> Because it brings in oxygen which keeps us alive	<u>Evolutionary history:</u> Because our first terrestrial ancestors developed lungs to adapt to life on land

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Sample statements from exam answers and media

- “territoriality evolved for the good of the species”
- “warning coloration helps to perpetuate the species”
- “without echolocation, the bat species would die out”



Sample statements from exam answers and media

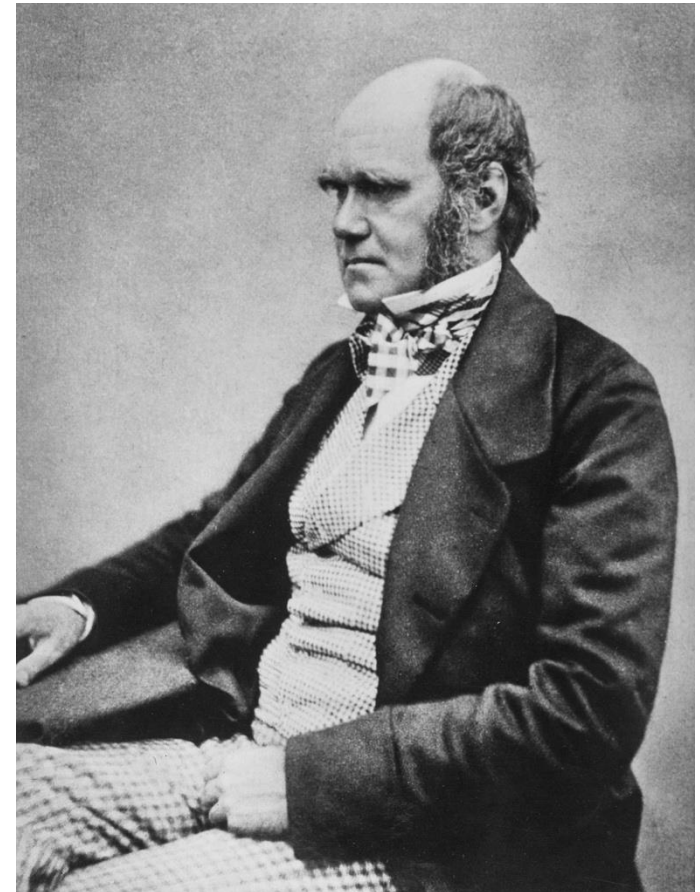
- Natural selection is purposeless and not acting for the good of anything



Darwin's theory of natural selection focused mainly on the level of the individual

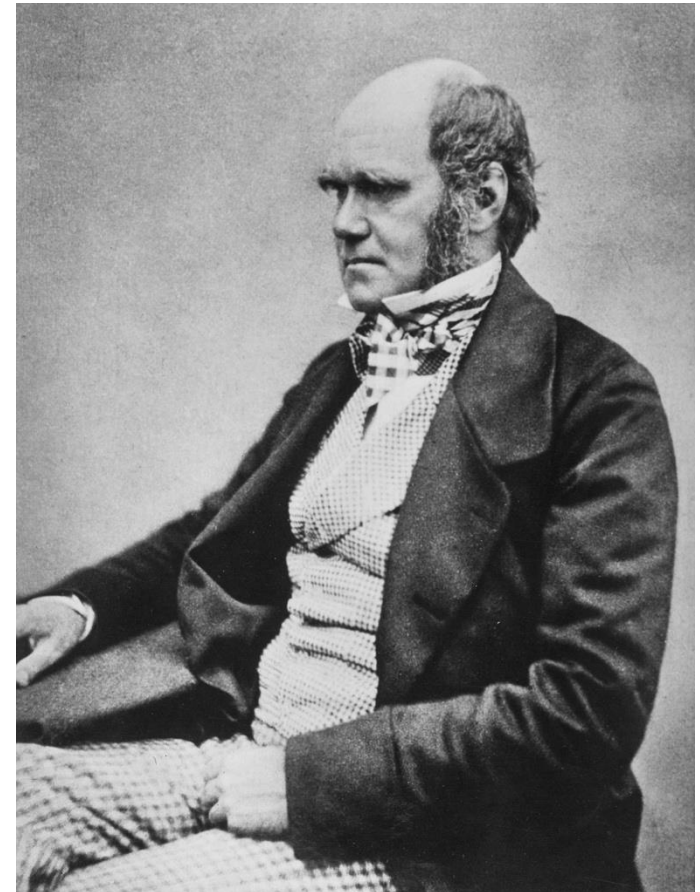
“...**individual** differences in the curvature or length of the proboscis...might profit a bee or other insect, so that certain **individuals** would be able to obtain their food more quickly (from flowers) than others....”

(Darwin, On the Origin of Species, 1859)



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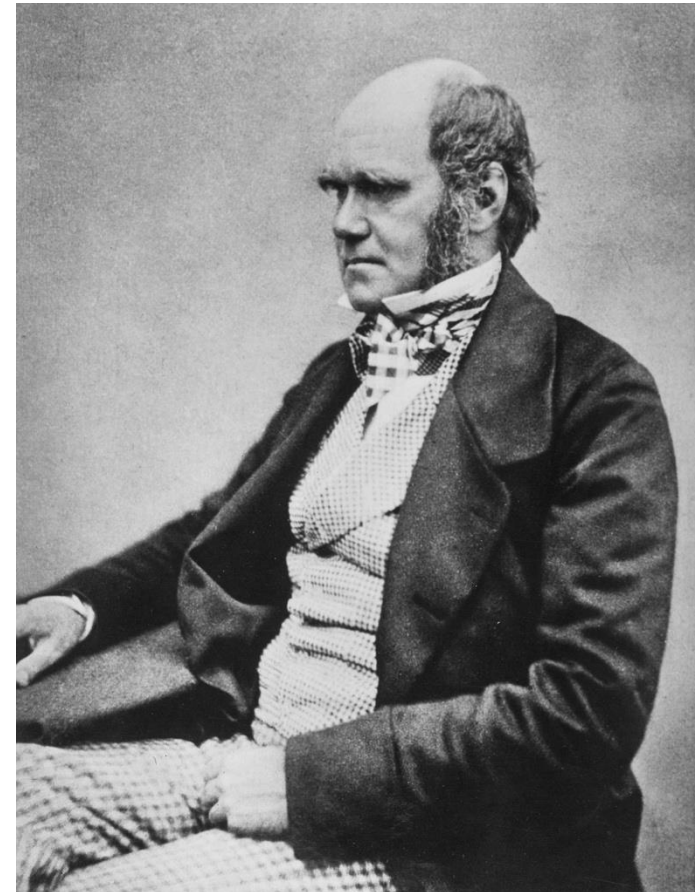
Why might we focus on selection at the level of the individual?



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Why might we focus on selection at the level of the individual?

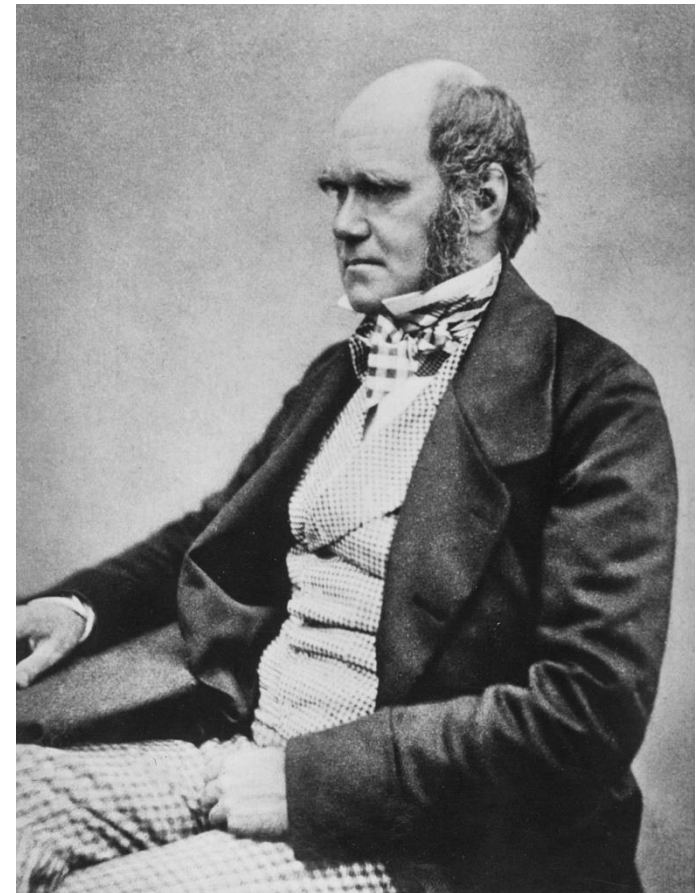
- Genetic information is passed on through individuals



Darwin's theory of natural selection focused mainly on the level of the individual

Why might we focus on selection at the level of the individual?

- Genetic information is passed on through individuals
- More on group selection and kin selection in the Cooperation lecture (Week 6)



Altruism in lemmings?

“Lemming suicide evolves at the level of the population, functioning to keep the populations from going extinct”
- Wynne-Edwards (1962)



This is a myth

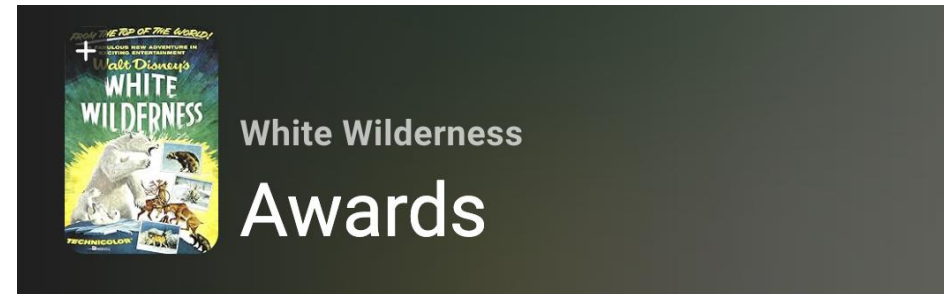
- Lemmings do NOT periodically hurl themselves into the sea
- Disney's video of lemmings allegedly jumping off a cliff was staged



White Wilderness, 1958

This is a myth

- Lemmings do NOT periodically hurl themselves into the sea
- Disney's video of lemmings allegedly jumping off a cliff was staged



[Jump to](#) ▼

2 WINS & 2 NOMINATIONS

Academy Awards, USA >



1959 Nominee Oscar

Best Music, Scoring of a Dramatic or Comedy Picture
[Oliver Wallace](#)



1959 Winner Oscar

Best Documentary, Features
[Ben Sharpsteen](#)

[James Algar](#) accepting this award.

This is a myth

- Lemmings do NOT periodically hurl themselves into the sea
- Disney's video of lemmings allegedly jumping off a cliff was staged
- **Cyclical explosions in population cause lemmings to migrate to areas of lesser density**



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How do we discuss scientific articles?

- Comprehension
 - What were the questions, hypothesis, etc.?
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 - What is the significance of the results?
 - How does it compare to past research?
 - Are there any ecological and/or evolutionary implications?
- Criticism
 - Was the study well-designed? What could they have done better? Were you convinced by their conclusions, or do you think further evidence or follow-up studies are needed?

Other possible directions

- Ask questions
- Imagine if you had to re-do and/or follow-up this study
- Build your own conclusions from the data, then compare them against the author's assertions in the discussion section
- Look at papers that cited this article

Testing function of fiddler **crab** claw waving by manipulating social context

DS **Pope** - Behavioral Ecology and Sociobiology, 2000 - Springer

... In this study, I experimentally manipulated the social context of male sand fiddler **crabs** (*Uca pugilator*) to investigate the effect of different audiences on the performance of the claw-...

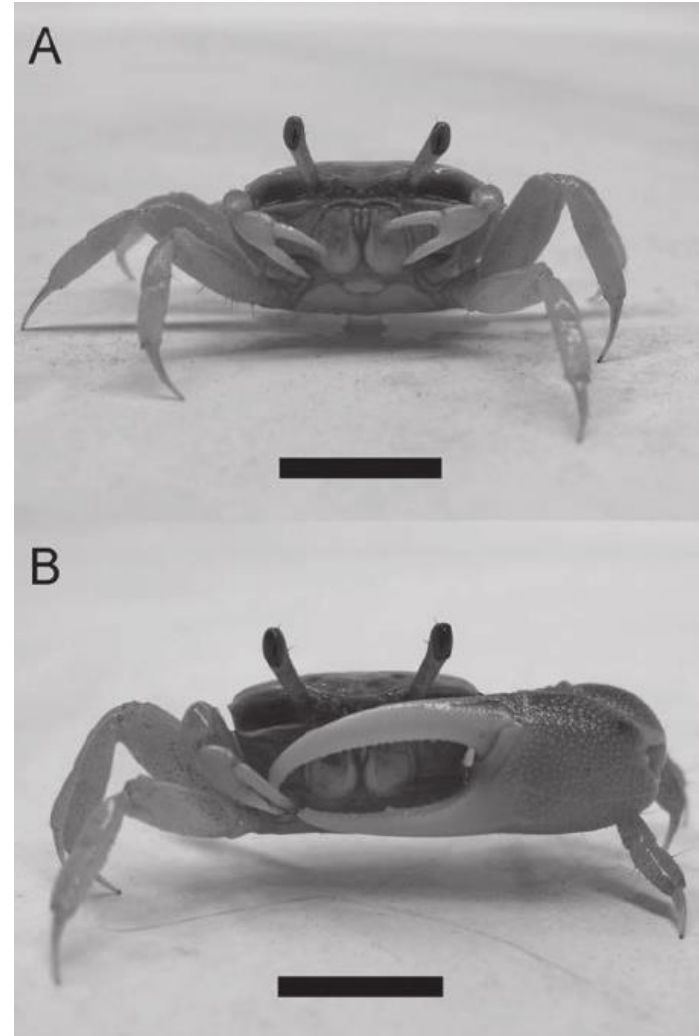
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Outline

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Atlantic sand fiddler crab
(*Uca pugilator*)



(A) Female and
(B) male fiddler crabs



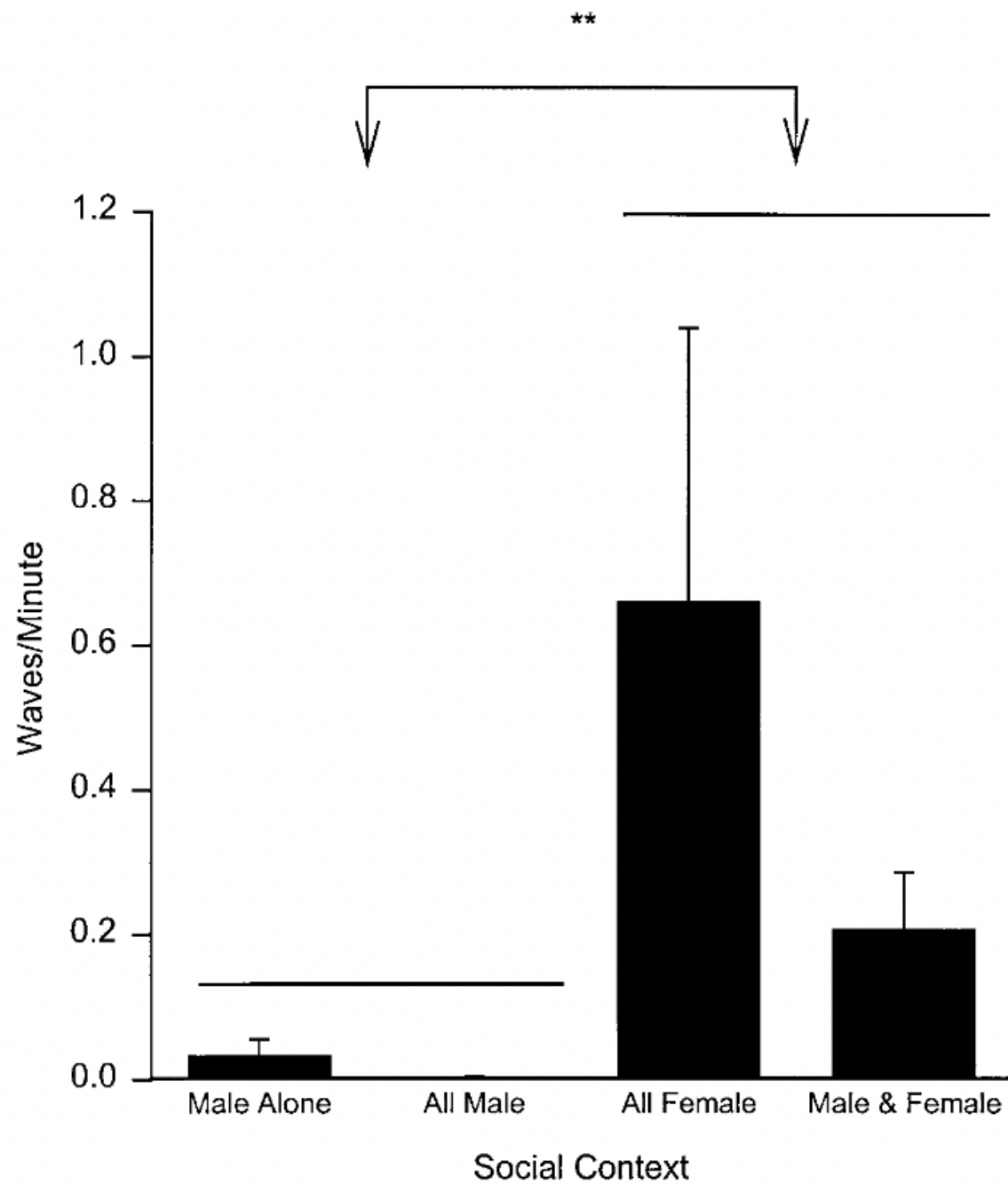


Fig. 1 Average waves/min of focal males in each social-context treatment (means+SE). There are significant differences ($P < 0.01$) between groups without females (male alone and all male) and treatments with females (all female and male and female). Sample size in each treatment is 15, except the all male ($n=14$)

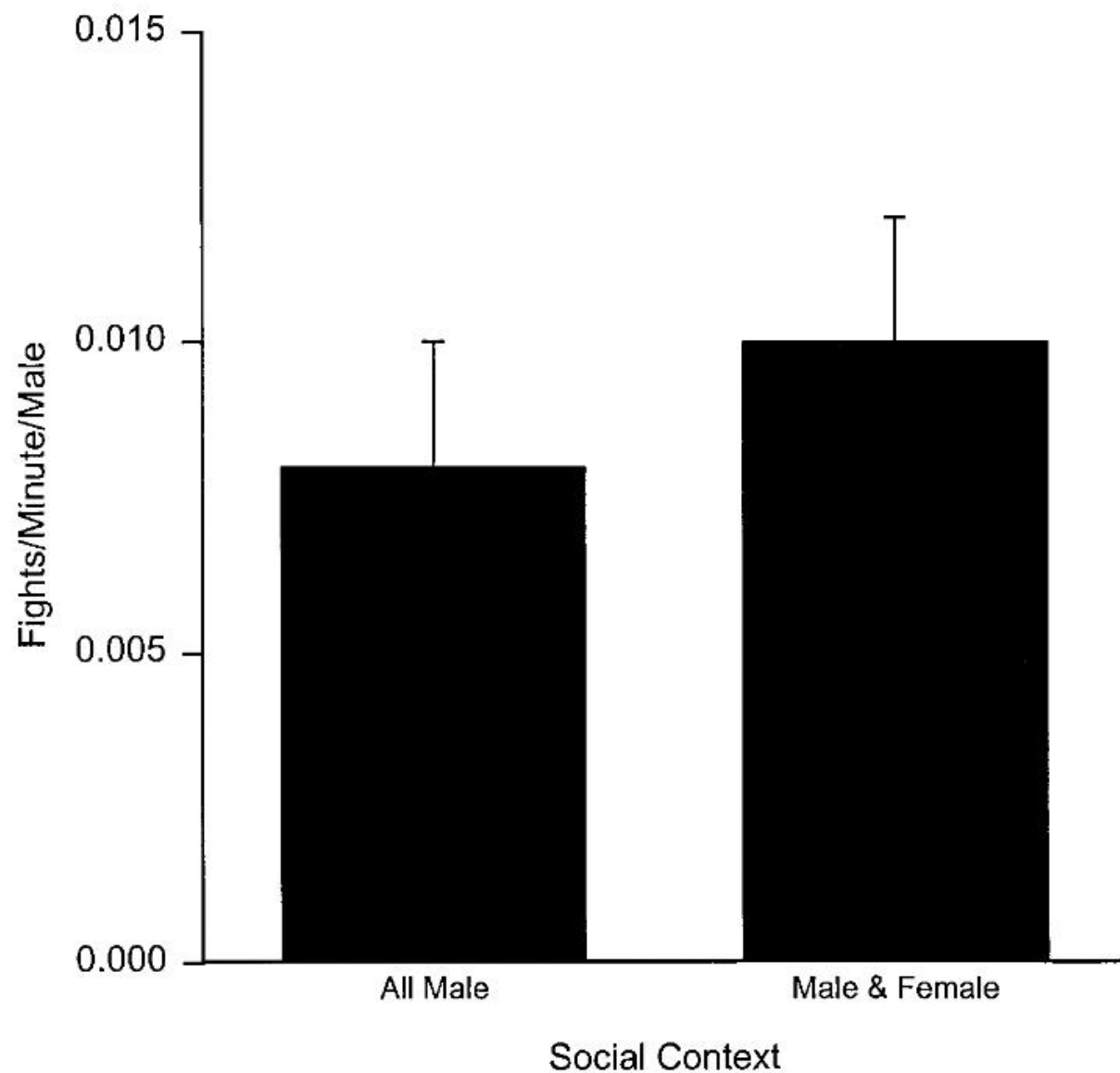


Fig. 2 Average (means+SE) fights/min per male of focal males in the two social-context treatments that included multiple males (all male and male and female). There is no significant difference between treatments ($P>0.5$)